



Indian Institute of Science Education and Research (IISER)

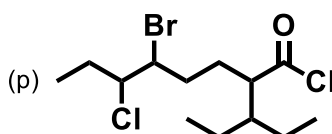
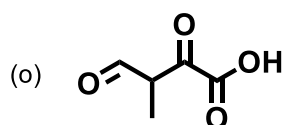
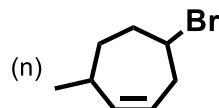
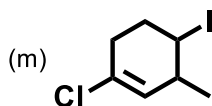
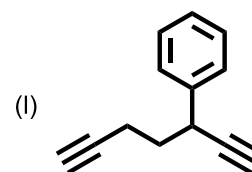
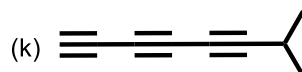
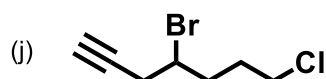
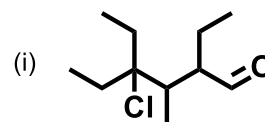
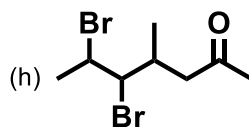
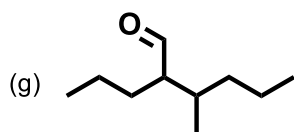
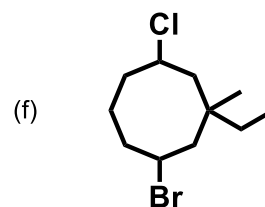
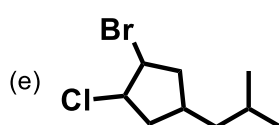
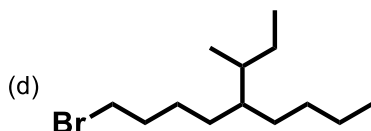
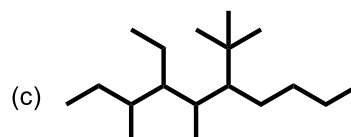
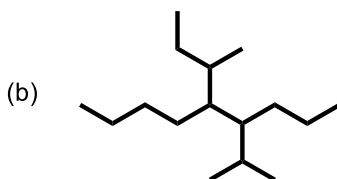
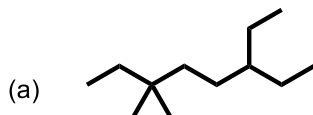
Mohali

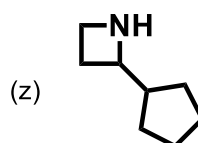
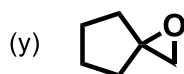
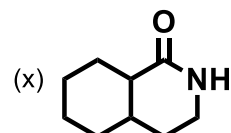
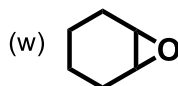
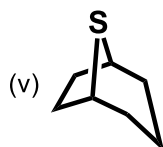
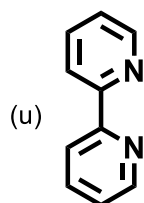
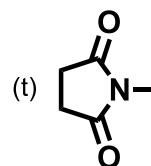
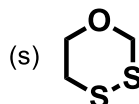
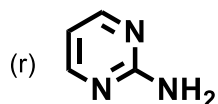
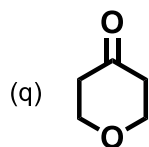
Sector 81, Knowledge City, S A S Nagar, Mohali, Punjab-140 036

CHM 102 Basic Organic Chemistry

Assignment - 1 (Nomenclature / Bonding concepts / Properties) – Spring 2024

Q1. Give the IUPAC name of the following structures:

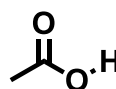




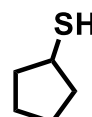
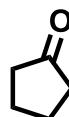
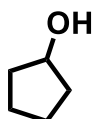
Q2. Draw the structure for compounds having IUPAC names as follows:

- 2-Ethyl-4-methyl-3-vinylpent-3-en-1-ol
- 3-Allyl-3-ethyl-1-methylcyclohexan-1-ol
- 1,2,3,4-Tetrahydropyrazine
- 4-(Pent-4-en-2-yn-1-yl)cyclohex-1-ene
- Thietan-2-one
- 4-Chloro-5-fluoro-1,2-oxazole

Q3. Draw the significant bond dipoles and molecular dipoles for the following molecules:



Q4. Identify the trend in the boiling points among the following compounds and indicate the reason.



Q5. Draw the frontier molecular orbitals for the following molecules:

- Cyclobutadiene
- 1,3,5,7-Octatetraene